

What is an IP Address?

An IP address (Internet Protocol address) is a unique numerical label assigned to each device connected to a computer network that uses the Internet Protocol for communication. It acts as an identifier for the device and enables it to communicate with other devices on the network.

An IP address has two main functions: to identify the host (device), and to provide the location of the host in the network. IP addresses are binary numbers, but they are most often written in human-readable notation, such as 192.168.0.1.

There are two versions of IP addresses in use today: IPv4 and IPv6. IPv4 addresses are 32-bit numbers and are written as four decimal numbers separated by dots, such as 192.168.0.1. IPv6 addresses are 128-bit numbers and are written as eight groups of hexadecimal numbers separated by colons, such as 2001:0db8:85a3:0000:0000:8a2e:0370:7334.

IP addresses are critical to the functioning of the internet and allow devices to communicate with each other and access the internet. When you connect to the internet, your device is assigned an IP address by your Internet Service Provider (ISP). The IP address is used to route traffic between your device and other devices on the internet.